Sarah Severinghaus

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EDUCATION

Reed College **August 2023 - May 2027**

Bachelor of Arts, Major in Physics, Minor in Mathematics *Portland, OR*

Honors: Commended for Excellence in Scholarship 2023-2024, 2024-2025, 2025-2026

Physics/Math Coursework: Physics 367 Computational Methods for Physics, Physics 366 Elementary Particles, Physics 342 Quantum Mechanics I, Physics 331/332 Advanced Laboratory I/II, Physics 311 Classical Mechanics I, Physics 321 Electrodynamics I, Physics 202 Modern Physics, Physics 211 Oscillations and Waves, Physics 231 Introduction to Electronics, Math 311 Complex Analysis, Math 322 Ordinary Differential Equations, Math 202 Vector Calculus, Math 201 Linear Algebra.

EXPERIENCE

Reed College **May 2026 - August 2026**

Research Fellowship, Physics *Portland, OR*

• Analytically and numerically investigated a system of identical, interacting quantum particles.
• Contributed to the manuscript detailing our work (submitted to Am. J. Phys.).
• Developed exercise sets for the Partnership for Integration of Computation into Undergraduate Physics (PICUP).

Reed College **August 2026 - Present**

Grader, Physics 321 - Electrodynamics *Portland, OR*

• Grades and gives feedback on homework assignments.

Reed College **October 2023 - Present**

Educational Technology Consultant, DataLab *Portland, OR*

• Supports students, faculty, and staff with Python, R, QGIS, and Excel.
• Develops educational resources and course content and assists faculty with data-intensive lab instruction.

University of Utah **May 2025 - August 2025**

Undergraduate Researcher *Salt Lake City, UT*

• Simulated cloud droplet formation in a cloud chamber with a one dimensional turbulence model.
• Processed and visualized simulation outputs in Python.
• Gained experience with high performance computing, atmospheric modeling, and cloud microphysics.

Reed College **August 2024 - December 2024**

Lab Teaching Assistant and Grader, Physics 164 - Stars and Stellar Systems *Portland, OR*

• Assisted students with data analysis and visualization in Python.
• Independently instructed telescope observation labs.
• Graded and provided feedback on homework assignments.

SKILLS

Languages: Python, R, C/C++

Packages/Frameworks: NumPy, Xarray, Pandas, Scikit-learn, Shiny, SciPy, Matplotlib, Git

Software: Mathematica, VSCode, LabVIEW, Arduino IDE, Jupyter, R Studio, QGIS